

Lookout: Hidden Camera Finder

Complete End-to-End Project Reference & Technical Dossier

Master Knowledge Base for Project Understanding, Research, and Google NotebookLM Context Ingestion

Project Name: Lookout (Hidden Camera Finder)	Target Event: iQOO City Battles, Hyderabad (26–27 Sept 2026)
Framework Architecture: React Native 0.86, Expo SDK 57/51 (Hermes)	Hardware Sensing: Collinear Optical Transceiver + 3D Magnetometer
Target Repository: Lookout---IQOO-Hyd- (Branch: main)	Validation Status: 100% Offline • 97/97 Automated Tests Passing

1. Executive Summary & Mission

Lookout is an offline-only mobile security application designed to detect covert hidden cameras (pinhole lenses, spy cameras disguised in smoke detectors, alarm clocks, tissue boxes, wall outlets) using standard, unmodified smartphone sensors and optics.

Core Tenet: Real Physics, Zero Gimmicks

Most commercial "camera detector" applications on mobile app stores display animated green radar graphics, play random audio beeps, and output arbitrary confidence scores (e.g., "98% Safe!"). Others claim to use "Cloud AI," requiring users to take photos of their private bedrooms or bathrooms and transmit them to external third-party servers.

Lookout is engineered on verifiable physical and mathematical principles:

- Optical Retroreflection:** Camera lenses are curved optical systems that reflect incident light straight back along the entry ray vector (the "cat's-eye" phenomenon).
- Collinear Optical Transceiver:** By leveraging the fact that a smartphone's flashlight/torch is physically mounted adjacent to its camera sensor (3–8 mm apart), the phone functions as a zero-parallax optical transceiver.
- Temporal Persistence Tracking:** A single image frame cannot distinguish a curved glass camera lens from a flat shiny screw head or metallic sticker. Lookout tracks candidate glints across consecutive frames as the user moves the phone. Flat specular surfaces flash for 1–2 frames and vanish; retroreflective lenses stay visible across multiple angles.
- Ferrous Magnetic Anomaly Detection:** Unshielded internal camera electronics, coils, transformers, and circuitry perturb the local ambient geomagnetic field, functioning as a non-verdict pre-filter to guide physical inspection.
- Air-Gapped Privacy:** Zero network code, zero external APIs, zero telemetry, and internet permissions programmatically blocked in the Android manifest. Operates 100% in Airplane Mode.

2. Problem Statement & Real-World Threat Landscape

2.1 The Covert Surveillance Epidemic

Covert surveillance has escalated rapidly across India and worldwide. Pinhole spy cameras are frequently uncovered in:

- Budget hotels, boutique guest rooms, homestays, and Airbnb rentals
- Retail trial rooms and fitting stalls in clothing stores and shopping complexes
- Women’s hostels, paying guest (PG) accommodations, and shared apartments
- Gym locker rooms, executive restrooms, and co-working spaces

Victims face extortion, non-consensual imagery distribution, and long-term psychological distress. Modern pinhole apertures measure between **1 mm and 2 mm** in diameter, making them virtually impossible for the human eye to detect under ambient room lighting (>95% failure rate in manual inspections).

2.2 The Fatal Flaw of Commercial Bug Detectors

- **RF Spectrum Detectors:** Traditional electronic bug detectors scan for radio frequency (Wi-Fi, Bluetooth, cellular) transmissions.
- **The Critical Indian Market Reality:** Over 90% of covert micro-cameras deployed in India record locally onto a hidden MicroSD card. They contain no Wi-Fi chips, emit zero radio frequency signals, and transmit nothing over the air. RF detectors and Wi-Fi scanner apps are 100% blind to offline SD-card cameras.
- **Dedicated Optical Wands:** Professional optical lens finders cost ₹3,000 to ₹15,000 (\$40–\$200), are bulky, require dedicated battery maintenance, and are rarely carried by everyday travelers.

2.3 The Failure of Existing App Store Software

- **The Placebo Problem:** Free apps on Google Play frequently output randomized confidence numbers, creating a deadly false sense of security.
- **The Cloud Privacy Hazard:** Apps requiring camera uploads to remote cloud servers defeat the very purpose of privacy, creating serious data breach and privacy violation risks.
- **Single-Frame False Positives:** Apps that attempt simple brightness thresholding on single frames flag every screw, mirror edge, door hinge, and glossy tile, triggering alert fatigue.

3. Physical Principles & Scientific Foundations

3.1 Principle 1: Optical Cat's-Eye Retroreflection

The Optical Principle: A camera lens is a retroreflective optical system consisting of multiple convex/concave glass elements and an aperture stop located near the focal plane. When collimated or semi-collimated light enters the entrance pupil, it is converged onto the sensor surface, where a portion is reflected back out through the same optical train, emerging along the exact reverse vector of the incident light.

Collinear Geometry: For retroreflected light to return to an observer, the illumination source must be co-axial with the sensor. Because smartphone manufacturers mount the rear LED flash directly adjacent to the camera sensor (within millimeters), the smartphone inherently functions as a collinear retroreflective transceiver without requiring auxiliary optical attachments.

Angular Envelope Dynamics:

- **Retroreflector (Lens):** Reflects incident light back to the source across an angular envelope of approximately $\pm 15^\circ$ to $\pm 30^\circ$. As the user moves the phone smoothly across a room, the returned glint remains visible over several consecutive frames.
- **Specular Reflector (Screws, Polished Tile, Stickers, Metallic Chrome):** Governed strictly by Snell's law ($\theta_r = \theta_i$). Incident light bounces off in a single directional ray. Unless the phone is positioned at the exact orthogonal angle relative to the flat surface, light does not return to the camera. During a hand sweep, specular reflections appear as transient 1-frame flashes and immediately vanish.

3.2 Principle 2: Ferrous Electromagnetic Perturbation

The Magnetic Principle: Pinhole cameras, micro-DVRs, clock radios, and disguised surveillance devices contain ferromagnetic components: power transformers, inductors, micro-speakers, PCB traces, and shielding cans.

Ambient Geomagnetic Baseline: The Earth's ambient magnetic field typically measures between $30\ \mu\text{T}$ and $60\ \mu\text{T}$ in residential environments.

Anomaly Detection: When a phone's 3D magnetometer passes within 2 to 10 cm of a concealed camera, the ferromagnetic core perturbs the local ambient field lines, producing a measurable deviation above the resting baseline.

Pre-Filter Role: Ferrous metal detection is strictly a *corroborating pre-filter*. It tells the user where to focus their optical lens scan, but cannot verify a camera on its own (since harmless wall screws, nails, and electrical wiring also produce magnetic deflections).

3.3 Principle 3: Infrared Emission & Hardware Constraints

The IR Principle: Many night-vision cameras illuminate their surroundings using invisible 850 nm or 940 nm Infrared (IR) LEDs.

Camera Sensor Response: Silicon CMOS sensors are naturally sensitive to near-infrared light, rendering IR illumination as a faint purple or white glow.

The Smartphone Reality (IR-Cut Filters): The primary camera on most modern flagship smartphones is equipped with an integrated hardware **IR-cut filter** designed to block infrared wavelengths to preserve daylight color accuracy. Front-facing selfie cameras occasionally omit this filter.

Engineering Decision: Lookout includes an Infrared detection mode (`DARK`), but ships it **disabled by default** (`IR_ENABLED_ON_THIS_DEVICE = false`). The user runs a 10-minute diagnostic protocol (`DEVICE_TESTS.md`) using a TV remote. If the camera sees IR, the flag is enabled; if blocked, the mode remains honestly grayed out with a clear explanation rather than faking support.

4. End-to-End Mathematical Algorithms ([src/logic/](#))



4.1 Computer Vision Pipeline ([blobDetect.js](#))

1. **Downscaling:** Frames are downscaled to 160×120 pixels (19,200 pixels). This allows pure JS execution in <8 ms while preserving the 3–30 pixel envelope of a pinhole lens at 0.5–2.0 meters.

2. **Luminosity:**

$$Y = 0.299 \times R + 0.587 \times G + 0.114 \times B$$

3. **Top-1.5% Adaptive Thresholding:** Dynamically calculates cutoff isolating top 1.5% brightest pixels (`BRIGHT_TOP_FRACTION = 0.015`) with an absolute floor `MIN_ABSOLUTE_BRIGHTNESS = 200/255` to block dark room sensor noise.

4. **8-Connected Flood Fill:** Clusters pixels meeting threshold into contiguous components, computing area A , bounding box, centroid $(\bar{x}, \bar{y})$, and peak brightness.
5. **Four Geometric Pinhole Filters:**
- **Area ($3 \leq A \leq 250$ px):** Discards single-pixel hot noise (<3 px) and lamps/windows (>250 px).
 - **Fill Ratio (≥ 0.5):** $\text{Area} / (\text{Width} \times \text{Height})$. True circular disc is $\pi/4 \approx 0.785$. Rejecting <0.5 eliminates diagonal streaks and scratches.
 - **Aspect Ratio (≥ 0.5):** $\min(W, H) / \max(W, H)$. Rejects reflections elongated along rods, wires, and edges.
 - **Local Contrast (≥ 60):** Peak brightness must exceed frame mean intensity by at least 60 units.

4.2 Multi-Frame Temporal Persistence Tracking (tracker.js)

- **Centroid Matching:** Candidates in frame *t* are matched to active tracks from frame *t-1* using nearest-neighbor Euclidean distance (≤ 14 px).
- **Persistence Counter:** Matched tracks increment persistence: $Persistence_t = Persistence_{t-1} + 1$.
- **Miss Tolerance:** Unmatched tracks survive up to 2 missed frames (`TRACK_MAX_MISSES = 2`) to tolerate minor frame drops.
- **Strong Track Escalation:** When a track reaches ≥ 3 consecutive frames (`STRONG_PERSISTENCE_FRAMES = 3`), it is promoted to a **Strong Track** (RED). Transient reflections from screws and stickers flash for 1–2 frames and reset.

4.3 3D Magnetometer Signal Processing (magnetometer.js)

- **Euclidean Magnitude:**

$$|B| = \sqrt{B_x^2 + B_y^2 + B_z^2} \text{ } \mu T$$

- **Rolling Baseline Buffer:** 20-sample circular buffer (2.0-second window) computes ambient baseline $\mu_{baseline}$ with a 20-sample warmup guard.
- **Relative Spike Detection:** Triggers when deviation $(|B| - \mu_{baseline}) / \mu_{baseline} \geq 0.15$ (15% rise above baseline).
- **Sustain & Debounce:** Requires 3 consecutive over-threshold samples (300 ms) to discard noise. Debounces with a 2500 ms cooldown to prevent log spam.
- **Haptics:** Tactile pulse frequency rises proportionally to magnetic magnitude.

4.4 Spatio-Temporal Sensor Fusion (fusion.js)

Because smartphones lack spatial positioning sensors, temporal proximity serves as a spatial proxy: an optical glint observed within a **4000 ms window** of a magnetic anomaly spike is fused into a single corroborated event.

4.5 The Verdict Decision Rules (verdict.js)

Observed Physical Signal	Verdict	User-Facing Headline	Description
Strong Glint (≥ 3 consecutive frames)	RED	Inspect this spot closely	Strong optical retroreflection detected.
Brief Glint + Correlated Metal Spike	RED	Inspect this spot closely	Optical reflection confirmed by local metal anomaly.
Brief Glint Alone (1–2 frames)	YELLOW	Possible lens reflection	Transient specular flash; worth verification.
Magnetic Spike Alone	YELLOW	Metal object, worth a lens check	Ferrous metal detected; no optical glint.
Infrared Glow Alone (if enabled)	YELLOW	IR glow detected	Infrared light source detected in darkness.
Clean Baseline (No anomalies)	GREEN	Clear	No physical anomalies detected during sweep.

The Four Inviolable Verdict Rules

1. **Rule 1: Strong Glint Alone CAN Reach RED.** Cheap micro-cameras in plastic casings have no metal shielding. If magnetic confirmation were mandatory, the app would be blind to plastic cameras.
2. **Rule 2: Magnetic Spike Alone Can NEVER Reach RED.** Hotel walls are full of harmless screws, hinges, and conduits. Classifying metal alone as a camera would trigger constant false alarms.
3. **Rule 3: Brief Glint + Metal Escalates to RED.** Two independent physical anomalies at the same location corroborate each other.
4. **Rule 4: GREEN Means "Clear", Never "Safe".** Absence of detection is not proof of absence. The word "safe" is programmatically banned.

5. Complete Configuration Reference (`src/config.js`)

Every tuning constant in Lookout is centralized in `src/config.js`:

Module	Constant	Value	Functional Role
MAGNETOMETER	<code>UPDATE_INTERVAL_MS</code>	<code>100</code>	Sensor sampling delivery period (10 Hz).
	<code>BASELINE_SIZE</code>	<code>20</code>	Rolling baseline sample buffer (2.0s rolling memory).
	<code>WARMUP_SAMPLES</code>	<code>20</code>	Initial samples discarded to establish stable ambient baseline.
	<code>SPIKE_RATIO</code>	<code>0.15</code>	15% magnitude increase above rolling baseline flags a spike.
	<code>SPIKE_SUSTAIN_SAMPLES</code>	<code>3</code>	Requires 3 consecutive over-threshold samples (300ms) to eliminate twitching.
	<code>COOLDOWN_MS</code>	<code>2500</code>	2.5-second debounce cooldown after logging a spike.
	<code>DISPLAY_CEILING_RATIO</code>	<code>0.6</code>	60% field deflection maps to maximum UI meter and haptic pulse rate.
GLINT (CV)	<code>CAPTURE_INTERVAL_MS</code>	<code>300</code>	Frame capture interval (~3.3 FPS in still-capture mode).
	<code>PROCESS_WIDTH / HEIGHT</code>	<code>160 × 120</code>	Downscaled frame resolution (19,200 px) for sub-10ms JS computation.
	<code>CAPTURE_QUALITY</code>	<code>0.4</code>	JPEG compression quality (lower = faster decode, bright spots preserved).
	<code>BRIGHT_TOP_FRACTION</code>	<code>0.015</code>	Top 1.5% brightest pixels isolated for connected components.
	<code>MIN_ABSOLUTE_BRIGHTNESS</code>	<code>200</code>	Absolute floor (0–255) to prevent noise triggers in dark scenes.
	<code>MIN_BLOB_AREA</code>	<code>3</code>	Minimum blob size (rejects single-pixel sensor noise).
	<code>MAX_BLOB_AREA</code>	<code>250</code>	Maximum blob size (rejects ceiling lights, lamps, TV screens).
	<code>MIN_FILL_RATIO</code>	<code>0.5</code>	Area / (W × H) (rejects diagonal scratches and glare lines).
	<code>MIN_ASPECT_RATIO</code>	<code>0.5</code>	min(W,H) / max(W,H) (rejects elongated reflections from rods and wires).
	<code>MIN_CONTRAST</code>	<code>60</code>	Peak blob brightness must exceed frame mean by at least 60 units.
	<code>TRACK_MATCH_DISTANCE</code>	<code>14</code>	Maximum centroid distance (px) to associate tracks across frames.
	<code>STRONG_PERSISTENCE_FRAMES</code>	<code>3</code>	Consecutive frames held required to escalate to Strong Track (RED).
FUSION	<code>CORRELATION_WINDOW_MS</code>	<code>4000</code>	4.0-second spatio-temporal fusion window correlating glint with metal spike.
IR	<code>ENABLED_ON_THIS_DEVICE</code>	<code>false</code>	Gated on physical test in <code>DEVICE_TESTS.md</code> (handles IR-cut filters).
	<code>CAMERA</code>	<code>"front"</code>	"front" or "back" based on hardware test outcome.

6. Quality Assurance & Verification Suite

Lookout contains **97 automated test assertions** executed across 4 test harnesses. The pure mathematical logic executes directly in Node.js in **under 1.0 second**.

Suite Name	File Path	Tests	Scope & Coverage	Status
Physics & Ground Truth	<code>tools/selftest.mjs</code>	33	Vector math, synthetic disc vs. streak rejection, persistence, verdict rules	33/33 PASS
Automation Unit Suite	<code>tools/test_automation.mjs</code>	44	Vector math, flood fill, centroid accuracy, tracker lifecycle	44/44 PASS
End-to-End Integration	<code>tools/test_integration.mjs</code>	12	Clean room vs. pinhole clock sweeps, multi-spot session rollup	12/12 PASS
Regression & Safety Audit	<code>tools/test_regression.mjs</code>	8	Timestamp key collisions, white/black frame extremes, forbidden copy	8/8 PASS
TOTAL TEST SUITE	4 Test Scripts	97	Complete Mathematical & Algorithmic Core	97/97 PASS

Key Ground-Truth Assertions Executed

- **Window & Lamp Rejection:** Proves an over-sized bright region (>250 px) is rejected.
- **Streak & Scratch Rejection:** Proves a diagonal scratch with fill ratio < 0.5 is rejected.
- **Sticker vs. Lens Persistence:** Proves an intermittent 1-frame specular flash never accumulates persistence and never reaches RED.
- **Magnetic Safety Rule:** Proves a 100× baseline magnetic spike alone can NEVER evaluate to RED.
- **Zero Forbidden Copy:** Audits all copy combinations to guarantee "Safe", "100%", or fake accuracy scores never appear.
- **1000-Event Collision Lock:** Generates 1,000 log events in the exact same millisecond and verifies zero React key collisions.

7. Hardware Calibration Protocols (`DEVICE_TESTS.md`)

Executed before deploying on a new smartphone model:

Test 1: Infrared Sensitivity Test (10 minutes)

In a dim room, point an ordinary TV remote control directly at the camera lens (10 cm away). Press and hold any remote button. If a purple/white glow is visible on the front camera: set `IR.ENABLED_ON_THIS_DEVICE = true` and `IR.CAMERA = "front"`. If visible only on rear: set `IR.CAMERA = "back"`. If invisible on both (due to hardware IR-cut filter): leave `false`. The mode remains honestly disabled in the UI.

Test 2: Optical Glint Range & Persistence Test (5 minutes)

Activate torch. Aim at a test camera lens from 30 cm, backing away until the glint fades (typically 2.0–2.5 meters). Sweep side-to-side across the target lens, verifying that the glint persists across frames. Sweep across a shiny metallic sticker or screw head, verifying that the glint flashes once and vanishes.

Test 3: Magnetometer Baseline & Spike Calibration (5 minutes)

Walk through a quiet room away from large metal appliances and record resting drift (typically 42–48 μT). Bring the phone near a known metal object (door hinge, wall screw) and note the peak reading (e.g., 62 μT). Set `SPIKE_RATIO` so the threshold sits comfortably above resting drift but below the metal peak. Ensure the quiet room remains completely silent.

Test 4: Frame Rate Reality Check (2 minutes)

Expected throughput is 2.0 to 3.5 FPS in Expo Go still-capture mode. If throughput drops below 1.5 FPS, reduce resolution in `config.js` to 128 × 96 pixels.

Test 5: Airplane Mode Verification (1 minute)

Engage Airplane Mode via Android settings. Execute a full workflow (Home → Sweep → Results → Evidence inspection). Verify zero permission prompts and zero network errors.

8. Live Demonstration Protocol (`DEMO_SCRIPT.md`)

- [0:00 – 0:30] **Problem Framing:** Highlight hidden camera crisis in India and the critical blindspot: over 90% of spy cameras record locally to MicroSD cards, rendering RF scanners completely useless.
- [0:30 – 1:00] **Physics of Retroreflection:** Hold up phone with torch illuminated. Explain cat's-eye retroreflection and the smartphone's collinear transceiver advantage.
- [1:00 – 2:00] **Live Scan & Persistence Counter:** Sweep phone toward real test lens on table. Point to HUD persistence counter climbing (1 → 2 → 3 → 4 → 5). Immediately sweep across a shiny metallic sticker: show reflection flash for 1 frame and reset to 0. Sweep an empty wall: demonstrate total silence.
- [2:00 – 2:30] **The Magnetometer Pre-Filter:** Address the hard objection upfront: *"It detects ferrous metal, not cameras. In our code, a magnetic spike can never produce a RED verdict on its own."*
- [2:30 – 3:15] **Forensic Evidence Inspection:** Tap "Finish Sweep". Display actual captured frame with reticle overlay, centroid coordinates, area (28 px), fill ratio (0.74), peak brightness (248/255), and rejection log showing 3 rejected bright spots.
- [3:15 – 3:45] **Air-Gapped Privacy:** Pull down Android notification shade showing Airplane Mode icon. Reiterate zero network permissions.
- [3:45 – 4:00] **Close:** *"We do not certify rooms as safe. Lookout points you at the exact physical spots worth inspecting, backed by verifiable physical evidence, completely offline."*
- **Emergency Stage Recovery:** If hardware fails, execute `npm run selftest` on laptop to show 33 passing assertions against synthetic ground truth in 0.8s.

9. Technical FAQ & Defense Matrix ([JUDGE_NOTES.md](#))

Q1: "The magnetometer can't tell a camera from a wall screw."

Answer: Correct, and Lookout never claims it can. The magnetometer detects ferrous metal as a cheap, fast pre-filter to guide physical attention. In [src/logic/verdict.js](#), a magnetic spike alone is strictly locked to YELLOW and can never reach RED. The UI explicitly states: *"Metal object, worth a lens check here"*, never *"Camera detected"*.

Q2: "This already exists on the Play Store."

Answer: Existing free tools suffer from two critical flaws: (1) They are RF-only, missing offline SD-card spy cameras; (2) They offer zero evidence verification, displaying opaque "Camera Found" alerts. Lookout provides full forensic evidence: the raw frame, reticle coordinates, blob metrics, threshold used, and the count and reasons for rejected candidates.

Q3: "What is your measured accuracy percentage?"

Answer: Lookout deliberately does not publish an invented accuracy percentage. Measuring an honest accuracy number requires representative testing across thousands of real-world rooms with planted cameras. Publishing an ungrounded "98% accurate" figure would be dangerous, as a user experiencing a false negative based on false confidence is worse off than one given raw physical evidence.

Q4: "What about a cheap plastic camera with no metal?"

Answer: A cheap micro-camera in a plastic case has virtually no magnetic signature. That is precisely why Rule 1 allows a strong persistent optical glint alone to trigger a RED verdict. If magnetic confirmation were mandatory, the app would be systematically blind to plastic spy cameras.

Q5: "Why does this need to be a mobile phone app?"

Answer: (1) Laptops do not possess magnetometers; (2) Retroreflection requires the light source to be collinear with the camera optical axis (3–8 mm spacing on phones; impossible on laptops or separate flashlights); (3) The user must physically sweep a room at 11 PM using a device already in their pocket.

Q6: "How would you productionize the 2–3 FPS frame rate?"

Answer: Plain Expo Go does not expose low-level camera frame processor buffers, requiring still captures and JS JPEG decoding. To productionize, we swap in [react-native-vision-camera](#) with C++ frame processors running directly on the camera thread at 30 FPS. Crucially, every mathematical function in [src/logic/](#) is pure and survives that swap completely unchanged.

10. Production Roadmap & Future Engineering

Phase / Horizon	Milestone Focus	Technical Deliverables
Phase 1 (Complete)	Zero-Dependency MVP	Expo SDK 51/57, pure JS CV pipeline, 10 Hz magnetometer, 97 automated tests passing.
Phase 2 (Next 30 Days)	30 FPS Native Pipeline	Integrate <code>react-native-vision-camera</code> with C++/Rust frame processors; keep <code>src/logic/</code> unchanged.
Phase 3 (Q1 2027)	Edge TinyML Classifier	Quantized 1.8 MB MobileNet edge model running on candidate crops to classify pupil reflections vs. screw threads.
Phase 4 (Q2 2027)	AR Spatial Room Pins	ARCore / ARKit plane detection anchors persistent 3D red warning reticles to suspicious room coordinates.

Lookout: Hidden Camera Finder — Comprehensive End-to-End Project Documentation (Hyderabad 2026)